Project: Worked examples
Title: PCB trace width (IPC-2221 legacy)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The minimum width of a printed-circuit trace for a current and an allowed temperature rise, from the legacy IPC-2221 curve-fit, and the resistance, voltage drop and I²R loss of the trace as drawn: legacy IPC-2221 curve-fit, conservative for multilayer boards. External or internal layer; copper weight in oz/ft²; resistance at the trace's operating temperature (ambient plus rise). The IPC-2152 method (charts, plane and board-thickness factors) is not implemented.
Cross-section: I = k·ΔT^0.44·A^0.725, so A = [I/(k·ΔT^0.44)]^(1/0.725) with A in mil², k = 0.048 external, 0.024 internal Copper thickness: t = w_Cu·t_oz, t_oz = 1.378 mil per oz/ft² (35 µm, derived) Minimum width: w_min = A/t; utilisation u_w = w_min/w_act Resistivity at the operating temperature: ρ_T = ρ_20·[1 + α·(T_amb + ΔT − 20)] Trace resistance: R = ρ_T·L/(w_act·t); voltage drop ΔV = I·R; loss P = I²·R
Ref: IPC-2221 6 — Electrical properties: current-carrying capacity of printed-board conductors (legacy curve-fit)
Ref: Hughes, Electrical and Electronic Technology Ch. on DC circuits — Resistivity and temperature coefficient of resistance

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
I	Trace current	5	A	spec: Driver datasheet DD-12: 5 A continuous motor supply	0.01 … 35 A
dT	Allowed temperature rise above ambient	20	K	spec: Board design rule BDR-3: 20 K allowed rise on power rails	1 … 100 K
T_amb	Ambient temperature at the board	25	degC	spec: Environmental specification ES-2: 25 °C at the board	-40 … 125 degC
layer	Layer	External layer		drawing: PCB-LAY-014 rev B	any
Cu_oz	Finished copper weight in oz/ft² (0.5 to 6), not a thickness in µm: 17.5 µm is ½ oz, 35 µm is 1 oz, 70 µm is 2 oz, 105 µm is 3 oz	2		spec: Fabrication note FN-7: 2 oz/ft² finished copper on the outer layers	0.5 … 6
w_act	Trace width as drawn	2.5	mm	drawing: PCB-LAY-014 rev B	0.05 … 10 mm
L	Trace length	100	mm	drawing: PCB-LAY-014 rev B	1 … 1000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_ext	Curve-fit constant k, external layer	0.048		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.01 … 0.1
k_int	Curve-fit constant k, internal layer	0.024		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.005 … 0.05
n_dT	Exponent of the temperature rise	0.44		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.3 … 0.6
n_A	Exponent of the cross-section	0.725		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.5 … 1
t_oz	Copper thickness per oz/ft² (35 µm = 1.378 mil, derived)	1.378	mil	reference: record: IPC-2221 legacy curve-fit constants; 1.378 mil = 35 µm per oz/ft², derived	1.2 … 1.5 mil
rho_20	Copper resistivity at 20 °C (annealed copper, 1.724e-8 Ω·m)	0.01724	ohm*mm^2/m	reference: record: annealed copper 1.724e-8 Ω·m at 20 °C, derived physical constant	0.0165 … 0.0185 ohm*mm^2/m
alpha	Temperature coefficient of resistance of copper	0.00393	1/K	reference: record: annealed copper, 0.00393 per K at 20 °C, derived physical constant	0.0035 … 0.0042 1/K

3. Assumptions

A1. The curve-fit is the legacy IPC-2221 relation, drawn for a trace on an outer or inner layer with no plane or convective help, 1 to 3 oz copper, up to 35 A, rises of 10 to 100 K and widths to 10 mm; it is conservative for multilayer boards and is a screening of width, not a thermal analysis of the board. (IPC-2221 curve-fit (legacy); IPC-2152 is not implemented.)

A2. Copper thickness is the nominal 1.378 mil (35 µm) per oz/ft² of finished copper; plating build-up on outer layers and etch-back are not modelled. (Derived: 35 µm / 25.4 µm per mil.)

A3. Resistance uses the trace as drawn at the temperature ambient + rise, from the resistivity at 20 °C and a linear temperature coefficient; vias, pads, bends and skin effect are ignored. (Derived constants for annealed copper.)
A4. Steady direct current (or RMS current at low frequency); the rise is the allowed rise in the worst case. (Scope of the curve-fit.)

4. Working

[14] k = k_ext — IPC-2221 §6 (legacy)
= 0.048
= 0.048
Curve-fit constant k (external layer)

[15] I_n = I / (1 A)
= 5 A / 1 A
= 5
Current as a number of amperes

[16] dT_n = dT / (1 K)
= 20 K / 1 K
= 20
Temperature rise as a number of kelvin

[17] A_req = (I_n / (k * dT_n^n_dT))^(1 / n_A) * (1 mil^2) — IPC-2221 §6 (legacy)
= (5 / (0.048 × 20^0.44))^(1 / 0.725) × 1 mil^2
= 98.511 mil²
Minimum conductor cross-section

[18] t_Cu = Cu_oz * t_oz
= 2 × 1.378 mil
= 70.002 um
Copper thickness

[19] w_min = A_req / t_Cu
= 98.511 mil² / 70.002 um
= 0.9079 mm
Minimum trace width

[20] u_w = w_min / w_act
= 0.9079 mm / 2.5 mm
= 0.36316
Width utilisation (minimum / as drawn)

[21] T_trace = T_amb + dT
= 25 degC + 20 K
= 45 degC
Trace operating temperature

[22] rho_T = rho_20 * (1 + alpha * (T_trace - 20 degC))
= 0.01724 ohm·mm²/m × (1 + 0.00393 1/K × (45 degC - 20 degC))
= 0.018934 ohm·mm²/m
Copper resistivity at the operating temperature

[23] A_act = w_act * t_Cu
= 2.5 mm × 70.002 um
= 0.17501 mm²
Cross-section of the trace as drawn

[24] trace_R = rho_T * L / A_act
= 0.018934 ohm·mm²/m × 100 mm / 0.17501 mm²
= 10.819 mohm
Trace resistance at the operating temperature

[25] vdrop = I * trace_R
= 5 A × 10.819 mohm
= 54.095 mV

Voltage drop along the trace

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[26] trace_Ploss = I^2 * trace_R
= (5 A)^2 × 10.819 mohm
= 270.47 mW
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Power loss in the trace (I^2R)

5. Results

SYMBOL	VALUE	DESCRIPTION
A_req	98.511 mil²	Minimum cross-section
w_min	0.9079 mm	Minimum trace width
u_w	0.36316	Width utilisation
trace_R	10.819 mohm	Trace resistance
vdrop	54.095 mV	Voltage drop
trace_Ploss	270.47 mW	Power loss

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Trace width as drawn against the minimum for the rise	u_w = 0.36316	u_w ≤ 1	0.363	IPC-2221 §6 (legacy)

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: k_ext, k_int, n_dT, n_A, t_oz (IPC-2221 legacy curve-fit constants); rho_20, alpha (Copper resistivity constants). Enter each as one so its source, edition and licence travel with it.

8. Verification

```
Template elec.pcb-trace@1.0.2 Content hash ae6d5ceb1daa0957 Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```